Context-Aware IoT-Based Field Monitoring Fertilization Guidance Module for Sri Lankan Paddy Cultivation Decision Support System

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Project Proposal Report

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DECLARATION

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Paddy cultivation in Sri Lanka, particularly during the Yala season, often results in low profit yields due to several practical issues faced by farmers. Incorrect crop selection, excessive or insufficient use of fertilizers, late detection of crop diseases, and heavy reliance on traditional farming knowledge without real-time field data lead to inefficient cultivation practices. Many farmers apply fertilizers based on fixed schedules or experience rather than actual soil conditions, which increases cultivation cost and reduces the quality and quantity of harvest. Since paddy farming plays a vital role in Sri Lanka's food production and rural economy, improving productivity while minimizing unnecessary expenses is essential to maintain sustainable farmer income and national food security.

To address this problem, this research proposes the development of an IoT-based smart monitoring component designed specifically for Sri Lankan paddy cultivation during the Yala season. The system continuously measures important soil and environmental factors that influence fertilizer effectiveness and crop growth. These factors include soil moisture, soil temperature, soil pH level, and NPK values. Monitoring these parameters in real time allows the system to identify the actual nutrient requirement of the crop instead of relying on general recommendations.

The collected data is transmitted through IoT devices to a centralized dashboard where the values are analyzed, and alerts are generated to notify farmers about the correct timing and required amount of fertilization. By providing timely recommendations, the system helps to prevent over-fertilization, reduce fertilizer waste, and avoid crop damage caused by improper nutrient management.

The expected result of this research is optimized fertilizer usage, improved crop health, reduced cultivation cost, and increased profit margins for Sri Lankan paddy farmers. The proposed solution encourages data-driven decision making and supports modernizing traditional farming practices through affordable IoT technology.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
REST API	Representational State Transfer Application Programming Interface
DoA	Department of Agriculture Sri Lanka
DSS	Decision Support System
NPK	Nitrogen, Phosphorus, Potassium
SLIIT	Sri Lanka Institute of Information Technology
PWA	Progressive Web Application
IoT	Internet of Things
DCS	Department of Census and Statistics (Sri Lanka)

1. INTRODUCTION

1.1 Background

Paddy cultivation in Sri Lanka plays a vital role in ensuring national food security and supporting the rural economy, especially during the Yala cultivation season where water availability and soil conditions must be carefully managed. However, many farmers experience reduced profit yields due to incorrect crop management practices, excessive or insufficient fertilizer usage, delayed disease detection, and strong dependence on traditional farming knowledge. According to agricultural guidelines published by the Department of Agriculture Sri Lanka, improper fertilizer timing and poor nutrient management are major reasons for low productivity in irrigated paddy fields [14], [16]. Farmers often apply fertilizers based on fixed schedules without considering real-time soil conditions such as moisture level, temperature, and pH, which directly affect nutrient absorption and plant growth [15], [24]. As a result, cultivation cost increases while the quality and quantity of harvest decrease, creating financial difficulties for small and medium-scale farmers [22].

Several existing technological solutions have been introduced to improve farming efficiency. IoT-based monitoring systems using wireless sensors have been developed to measure soil moisture, temperature, and environmental conditions in real time, allowing better observation of field status [7], [12]. Some smart irrigation systems are capable of automatically controlling water supply in paddy fields, but they mainly focus on irrigation management rather than fertilizer timing [5], [6]. Decision-support systems have also been proposed to provide recommendations using collected data, yet these systems often require expensive equipment and technical knowledge, making them less suitable for rural farmers [10], [11]. Precision agriculture methods using sensor networks and cloud platforms have shown improvements in crop performance, but many of these solutions are not specifically designed for Sri Lankan paddy cultivation conditions [1], [9]. Therefore, a low-cost IoT-based monitoring component that focuses on soil factors and fertilizer timing is necessary to support accurate, real-time decision making for farmers.

1.2 Literature Survey

Paddy cultivation in Sri Lanka is a fundamental agricultural activity that supports national food security and rural livelihoods, particularly during the Yala cultivation season. Despite its importance, farmers frequently experience reduced productivity and profit margins due to inefficient fertilizer management, lack of real-time soil monitoring, and dependence on traditional knowledge. Research indicates that improper fertilizer application either excessive or insufficient, leads to nutrient imbalance, increased cultivation costs, and decreased yield quality [14], [16], [30]. Furthermore, the absence of continuous monitoring systems prevents farmers from adapting to dynamic soil and environmental conditions, limiting their ability to optimize

fertilization practices [22], [26].

Recent advancements in smart agriculture have introduced IoT-based monitoring systems that enable real-time data collection from agricultural fields. Studies have demonstrated that wireless sensor networks can monitor parameters such as soil moisture, temperature, and humidity, improving irrigation efficiency and general farm management [7], [12], [27]. Some implementations integrate cloud-based dashboards to visualize data and support decision-making processes [10], [28]. While these systems provide valuable insights into environmental conditions, they are primarily focused on irrigation control rather than nutrient management, leaving a critical gap in fertilizer optimization [5], [6].

Further research in precision agriculture highlights the importance of soil parameters such as pH and electrical conductivity in determining nutrient availability and fertilizer requirements [15], [24], [29]. Decision-support systems have been developed to recommend fertilization schedules; however, these systems often rely on static guidelines that do not adapt to real-time soil conditions, reducing their effectiveness in dynamic environments [1], [9]. Additionally, many existing solutions require expensive infrastructure and technical expertise, limiting their accessibility to small and medium-scale farmers in Sri Lanka [11], [22].

To address these limitations, this research proposes an IoT-based fertilization monitoring component that integrates real-time soil sensing with a decision-making model tailored to Sri Lankan paddy cultivation during the Yala season. The system continuously measures soil moisture, temperature, pH, electrical conductivity, and humidity, and generates alerts for optimal fertilizer timing and quantity. By combining sensor data with localized agricultural guidelines, the proposed solution enables data-driven fertilization decisions, reduces unnecessary fertilizer usage, and improves crop yield and profitability. This approach directly addresses the shortcomings of existing solutions by focusing specifically on fertilizer optimization rather than general farm monitoring.

After reviewing both academic studies and existing smart agriculture solutions, three limitations emerge that collectively justify the novelty of this component:

1. No existing system provides real-time, soil-driven fertilization recommendations tailored to Sri Lankan paddy cultivation. Most reviewed systems rely on fixed schedules or general guidelines rather than continuously measured soil parameters such as moisture, pH, and nutrient availability [13], [14], [16], [30]. This prevents farmers from making accurate, condition-based fertilization decisions at the right time.
2. No system translates collected soil data into actionable, timing-specific fertilization alerts that clearly indicate when to apply fertilizer, how much to apply, and the potential impact of delayed or excessive application. Existing solutions primarily provide raw data or general recommendations, not decision-oriented guidance that supports farmer actions [10], [11], [26], [27].

3. No system has been specifically designed and validated for Sri Lankan paddy fields during the Yala season, incorporating local soil variability, climate conditions, and fertilizer usage constraints such as import limitations. Most existing solutions are either generic precision agriculture tools or designed for other regions, making them less effective under Sri Lankan agricultural conditions [15], [20], [28], [29].

Author(s)	Institution / Source	Contribution	Key Limitation
K. Fernando, P. De Silva, and H. Wijesinghe (2020) [7]	IEEE IoT Agriculture Study	Developed IoT-based soil monitoring system for agriculture	Focuses on monitoring, lacks fertilization decision support
P. Rathnayake, S. Bandara, and T. Perera (2021) [5]	Smart Irrigation Research	Automated irrigation control using sensor data	Limited to water management, no nutrient optimization
A. Dissanayake, M. Hettiarachchi, and S. Perera (2023) [10]	Precision Agriculture Study	Cloud-based dashboard for farm data visualization	Does not provide real-time fertilization alerts
J. K. Nayak et al. (2020) [26]	IEEE IoT Journal	Comprehensive IoT agriculture framework and applications	Generalized approach, not specific to fertilization or rice crops
S. R. Mishra et al. (2021) [27]	IEEE Access	ML-based precision agriculture decision support systems	Lacks localized adaptation to Sri Lankan soil and climate conditions

1.3 Research Problem

Paddy farmers in Sri Lanka frequently experience reduced profit margins due to improper fertilizer management, poor monitoring of soil conditions, and reliance on traditional cultivation practices, especially during the Yala season. Many farmers apply fertilizers according to fixed schedules recommended in general guidelines, without considering real-time variations in soil moisture, pH level, temperature, and nutrient availability. According to agricultural studies, incorrect timing and excessive use of fertilizers not only increase cultivation cost but also reduce crop yield due to inefficient nutrient absorption by the plant [14], [16]. This problem is more noticeable in small and medium-scale farms where modern monitoring tools are not available and decisions are based mainly on past experience [22].

Existing research shows that real-time monitoring using IoT sensors can improve agricultural decision making by providing accurate field data [7], [12]. However, most currently available systems mainly focus on irrigation

control or general farm monitoring and do not specifically address fertilizer timing based on soil condition analysis [5], [6]. Studies on nutrient management in rice cultivation indicate that measuring soil factors such as moisture, pH, and temperature can help determine the correct fertilizer requirement at each growth stage [15], [24]. Therefore, there is a need for a low-cost IoT-based monitoring component that can continuously measure soil conditions and generate alerts to guide farmers on the correct timing of fertilization in Sri Lankan paddy cultivation.

1.4 Research Gap

Although significant advancements have been made in IoT-based agricultural monitoring and precision farming, several gaps remain in the context of Sri Lankan paddy cultivation, particularly during the Yala season. Existing solutions such as wireless sensor networks and IoT-based monitoring systems have been developed to measure soil moisture, temperature, and environmental conditions, providing data that can improve irrigation and general farm management [7], [12]. However, these systems are often designed for large-scale farms or research purposes, requiring high-cost equipment and technical expertise that is not accessible to most small and medium-scale Sri Lankan farmers [10], [11].

Another limitation of current implementations is that many smart farming systems focus primarily on irrigation control rather than fertilizer management [5], [6]. While decision-support systems can analyze collected data and provide recommendations, they rarely offer real-time, actionable alerts specifically for fertilizer timing [1], [9]. Furthermore, most of these systems do not monitor all key soil factors necessary for optimal nutrient application, such as soil pH, NPK, and ambient humidity, which directly affect nutrient absorption and plant growth [15], [24]. Agricultural guidelines suggest that proper fertilizer timing and precise application based on soil conditions can improve yield and reduce cultivation costs, yet farmers often lack tools to implement this in practice [14], [16].

The proposed IoT-based component aims to bridge these gaps by providing a low-cost, practical solution tailored for Sri Lankan paddy farmers. By continuously monitoring(daily/weekly) critical soil parameters including moisture, temperature, pH, NPK, and ambient humidity, the system will generate real-time alerts for the correct timing and amount of fertilizer application [7], [13], [24]. Unlike existing solutions, the component focuses specifically on fertilizer management, reducing waste, improving crop growth, and increasing profit margins for farmers while supporting data-driven decision-making in local farming conditions [10], [22].

2 OBJECTIVES

2.1 Main Objectives

The main objective is to develop an IoT-based smart monitoring component that helps Sri Lankan paddy farmers optimize fertilizer usage and improve crop yield during the Yala season. The system continuously measures(daily/weekly) critical soil parameters, including moisture, temperature, pH, NPK, and ambient humidity, to assess the field's nutrient requirements. By analyzing this real-time data, the component generates timely alerts on the optimal timing and quantity of fertilizer application. This approach reduces excessive or insufficient fertilizer use, lowers cultivation costs, enhances nutrient absorption, and supports data-driven decision-making, ultimately increasing the profitability and sustainability of paddy farming.

2.2 Specific Sub-Objectives

IoT monitoring devices are installed across multiple zones of the paddy field, including the beginning, middle, and end of the cultivation area. This setup captures spatial variations in soil conditions during the Yala season, ensuring that the collected data accurately reflects the overall field environment for informed fertilization decisions.

ESP32-based IoT devices continuously measure soil moisture, soil temperature, soil pH, electrical conductivity, and ambient humidity at fixed intervals. The readings are transmitted to a central server, providing the environmental data necessary to analyze and determine the optimal timing for fertilizer application.

A REST API–based data transmission system is combined with a PostgreSQL database to store sensor readings along with timestamps and zone identifiers. The collected data is processed by calculating both zone-specific and average soil values, creating reliable datasets for training the fertilization decision model.

Using the collected soil data alongside recommended Sri Lankan Yala season fertilization guidelines, the system determines the ideal timing and quantity of fertilizer application. Alerts are generated whenever soil conditions fall outside the required nutrient range, helping farmers take timely action.

A web-based dashboard displays real-time soil conditions and automatically notifies farmers about the correct fertilization timing. It also offers alternative fertilizer recommendations, such as organic or bio-fertilizers, when chemical fertilizers are limited due to logistical or import restrictions.

2.3 Anticipated Objectives & How we expect to solve

The success of the proposed component **depends on the availability of IoT sensors and microcontrollers**. While basic devices like soil moisture, temperature, and ambient humidity sensors are generally available in Sri Lanka, more specialized sensors such as pH probes and electrical conductivity meters may need to be imported. Considering the fact Yala season date from April-September, there must be a time conflict on getting the devices. To mitigate this, the system can initially use locally available sensors and historical soil data for preliminary testing, with advanced sensors integrated later to improve accuracy and model training.

Sri Lanka's dependency on imported chemical fertilizers makes paddy cultivation vulnerable to logistics disruptions caused by war and geopolitical instability. Delays in fertilizer import or restricted supply chains could prevent farmers from applying fertilizers at the recommended times, reducing crop yield and efficiency. The proposed system mitigates this by prioritizing precise and optimized fertilizer usage based on real-time soil data, ensuring minimal waste of available stock. It can also suggest alternative nutrient sources such as compost, bio-fertilizers, or green manure, allowing farmers to maintain crop growth and productivity even when conventional fertilizers are delayed or unavailable.

The current island-wide heat wave can affect both the accuracy of IoT sensors and the effectiveness of fertilizers. High temperatures accelerate soil drying, leading to frequent irrigation requirements, and can cause nutrient loss through volatilization if fertilizers are applied incorrectly. Sensors exposed to direct sunlight or high humidity may experience drift or degradation, impacting data quality. To address this, sensors must be protected using shielded housings and regularly calibrated. The system will also adjust fertilizer recommendations based on ambient temperature and soil conditions to maintain nutrient efficiency and optimize crop growth.

3 METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, combining iterative IoT system design, soil data collection, ML-based fertilization decision modeling, and dashboard engineering. DSR is appropriate because the primary output is a working artifact (the IoT fertilization monitoring component) whose effectiveness is evaluated against defined criteria such as prediction accuracy, alert timeliness, and farmer usability. The approach proceeds across five phases, each aligned with specific objectives of the component.

3.1 Research Approach

The research follows the DSR cycle: (1) Problem identification and motivation (literature review and gap analysis, Section 1.2–1.3), (2) Definition of objectives (Section 2), (3) Design and development (Phases 2–4 below), (4) Demonstration (field-deployed IoT devices and dashboard), and (5) Evaluation (model validation and farmer usability testing). This cycle ensures that the component design addresses the identified problem of inefficient fertilization and that evaluation metrics are pre-defined and measurable.

3.2 Phase 1 – Data Collection and IoT Device Setup

IoT devices built using ESP32 microcontrollers will integrate soil moisture, soil temperature, soil pH, electrical conductivity, NPK, and humidity sensors and will be deployed across multiple zones of selected paddy fields. Sensor data will be collected at fixed intervals and transmitted via WiFi or LoRa modules to the central server. Communication between devices and server will be handled using RESTful APIs over HTTP/HTTPS, with optional use of MQTT protocol for lightweight messaging. Incoming data will be stored in a PostgreSQL database, where preprocessing operations such as outlier removal (using Python libraries like Pandas), missing value imputation, and normalization will be applied. Historical datasets from DCS Sri Lanka and Hector Kobbekaduwa Institute will be integrated into the database for model training. Ethical approval for field data collection will be obtained from the SLIIT Research Ethics Committee.

3.3 Phase 2 – Fertilization Decision Model

The fertilization decision model will be developed using Python with Scikit-learn, supported by Pandas and NumPy for data preprocessing and feature engineering. The model will analyze processed sensor data and historical datasets to determine optimal fertilization timing and quantity. Features will include soil parameters, crop growth stages, and Yala season guidelines. Both rule-based logic and machine learning techniques will be implemented to improve prediction accuracy. Model evaluation will be conducted using metrics such as accuracy, precision, and R-squared, ensuring reliable recommendations. The trained model will be integrated into the backend using FastAPI, enabling real-time processing of incoming sensor data and generation of fertilization alerts.

3.4 Phase 3 – Dashboard and Alert System

A Next.js-based Progressive Web App (PWA) will be developed to provide farmers with real-time access to soil data and fertilization alerts. The dashboard interface will be designed using Tailwind CSS or Bootstrap for responsive layouts, while Chart.js will be used to visualize soil trends and alert data. The backend, built with FastAPI, will expose REST API endpoints to retrieve processed data, alerts, and historical records from the PostgreSQL database. Alerts generated by the decision model will be displayed in Sinhala and English using i18next for multilingual support, ensuring accessibility for local farmers.

3.5 Phase 4 – Deployment, Testing, and Evaluation

The complete system will be deployed on a Hostinger VPS, using Docker for containerization to ensure consistent performance across environments. Version control and collaboration will be managed using GitHub, enabling continuous integration and updates. Field testing will be conducted with farmers from selected districts, where system performance will be evaluated based on alert accuracy, response time, and usability. Real-time data flow, API reliability, and dashboard responsiveness will be monitored during testing. Based on evaluation results, adjustments will be made to sensor placement, model parameters, and system performance to ensure optimal functionality in real-world paddy field conditions.

3.6 Tools and Technologies

The proposed IoT-based paddy fertilization monitoring system utilizes a combination of hardware and software technologies to enable real-time data collection and intelligent decision-making. At the field level, IoT devices built using ESP32 or Arduino MKR WiFi microcontrollers are integrated with soil sensors to measure parameters such as soil moisture, temperature, pH, electrical conductivity, and ambient humidity. These devices are selected for their low power consumption, affordability, and compatibility with wireless communication protocols, making them suitable for deployment in Sri Lankan paddy fields, especially during the Yala season. For communication and data transfer, the system uses lightweight and reliable protocols such as RESTful APIs and MQTT to transmit sensor data from the field devices to the central server. Depending on field conditions, WiFi or long-range communication technologies such as LoRa can be used to ensure connectivity in remote agricultural areas. The backend system is developed using Node.js or FastAPI, providing a scalable and efficient platform for handling incoming data, processing requests, and managing system logic.

Data storage and processing are handled using PostgreSQL as the primary relational database, ensuring structured storage of sensor readings, timestamps, and field identifiers. For data analysis and model development, Python-based libraries such as Pandas, NumPy, and Scikit-learn are used to preprocess data and build the fertilization decision model. These tools enable efficient handling of time-series data, feature engineering, and the generation of accurate recommendations based on soil conditions.

The user interface is developed as a Progressive Web App using Next.js, allowing farmers to access the system through both mobile and desktop devices. Visualization libraries such as Chart.js are used to display real-time soil data, trends, and alerts in an intuitive format. The system is deployed on a Hostinger VPS, with GitHub used for version control and collaboration, ensuring maintainability, scalability, and continuous improvement of the application.

3.7 Validation Strategy and Performance Metrics

Validation will be conducted at three levels:

- **Model level:** R-squared and precision for fertilization decision model.
- **System level:** Real-time alerts back-tested against historical Yala season fertilization outcomes.
- **User level:** Dashboard usability with at least 10 farmers, aiming for 80% task completion rate.

3.8 Data Collection and Ethical Considerations

All historical datasets are publicly accessible. Farmer participation is voluntary with informed consent. No personally identifiable information is stored. Ethical approval will be sought from the SLIIT Research Ethics Committee before field data collection.

4 High Level Architecture Diagram

4.1 Overview of Proposed Solution

The proposed system is a four-module Smart Agriculture DSS for Sri Lankan Paddy Cultivation (Project R26-IT-066). Each module addresses a distinct problem domain and communicates with others via REST APIs through a shared cloud layer. The four modules and their relationships are:

- IT22178718 - Context-Aware IoT Field Monitoring: Continuously collects real-time soil and environmental sensor data (moisture, pH, NPK, temperature, humidity, water level). Sends processed sensor readings to this component (IT22266446) as inputs to the base yield estimation model.
- IT21811104 - Intelligent Crop Recommendation: Analyses IoT and climate data against market supply/demand forecasts. Sends current paddy market price per kg to this component for revenue projection calculations within the scenario engine.
- IT22261632 - Disease Detection and Severity Analysis: Detects Rice Blast and Brown Spot from leaf images and environmental sensor data. Estimates disease severity percentage and growth stage at detection. Sends disease type, severity score, and growth stage to this component as inputs to the disease-yield loss regression models.
- IT22266446 - Disease-Aware Harvest Yield Estimation: Receives IoT data, disease severity scores, and market price from the three other modules. Applies the base yield model and disease-yield loss models to generate three-scenario yield and revenue forecasts with net treatment benefit calculation. Delivers outputs via REST API to the unified group dashboard.

4.2 Individual Component Architecture

The **IoT-Based Fertilization Monitoring Component** consists of four internal processing layers, each directly aligned with a specific sub-objectives:

Layer 1 – Sensor Data Ingestion and Validation (SO1, SO2)

Receives real-time soil data (moisture, temperature, pH, electrical conductivity, humidity) from field-deployed ESP32/Arduino IoT devices at configurable intervals (default: 15 minutes). Data is collected across multiple zones of the paddy field and aggregated using zone averages. Applies preprocessing including outlier detection (IQR method), missing value imputation (median for sensor, forward-fill for time-series), and normalization before passing validated data to the fertilization decision engine. Optional integration of historical yield and market data supplements model training.

Layer 2 – Fertilization Decision Engine (SO2, SO4)

Processes validated soil sensor readings using a rule-based and ML model to determine optimal fertilization timing and quantity. Growth stage (germination, tillering, panicle initiation, heading, grain filling) and Yala season guidelines are included as features. Generates soil nutrient threshold alerts and calculates expected fertilizer dose. Incorporates alternative fertilizer logic for chemical supply shortages. Outputs actionable recommendations for field-specific zones.

Layer 3 – Revenue and Benefit Calculator (SO4, SO5)

Combines predicted fertilization schedules with current market prices (Hector Kobbekaduwa datasets) to estimate cost savings and potential revenue improvement per field zone. Retrieves local treatment costs from system database references. Computes net benefit of recommended fertilization strategies and generates clear decision indicators (Apply / Do Not Apply) with financial justification for farmer guidance.

Layer 4 – Dashboard and REST API Layer (SO4, SO5)

Exposes forecast outputs and alerts via authenticated FastAPI endpoints. A React.js dashboard renders real-time soil data trends, recommended fertilizer quantities, alternative options, and net benefit visualizations in Sinhala and English. Dashboard auto-refreshes upon receiving new IoT or historical data and provides an interactive, user-friendly interface for smallholder farmers to make informed fertilization decisions.

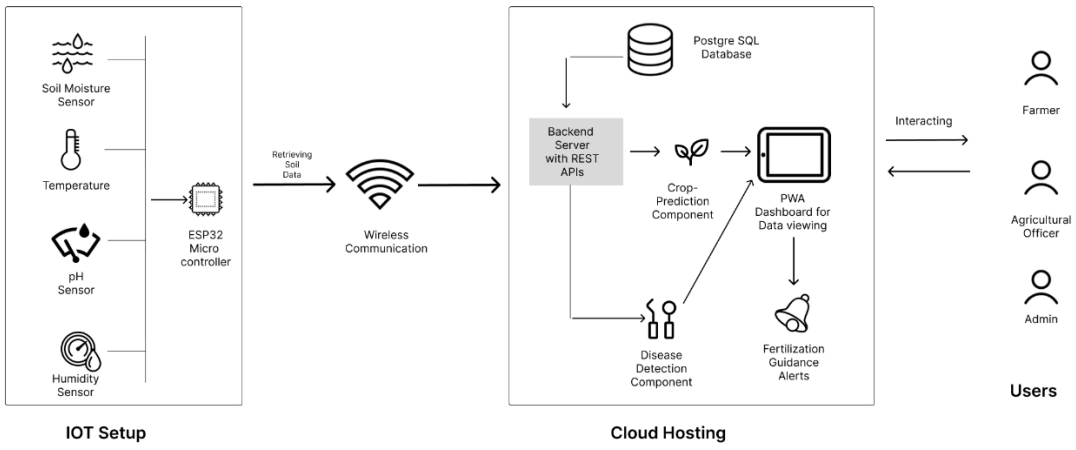


Figure 1 -- System Architecture

5 USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Users: Small and medium scale paddy farmers in Sri Lanka (Main focus – Kurunegala district)
- Secondary Users: Agricultural officers and agronomists who analyze data and provide cultivation recommendations.
- System Consumers: IT2226162 depends on my component data to analyze disease severity.
- Administrative Users: System administrators managing model training and managing user accounts.

5.2 Functional requirements

ID	Requirement
FR-01	The system shall receive real-time IoT sensor data including soil moisture, soil temperature, soil pH, electrical conductivity, and ambient humidity from field devices via an authenticated REST API endpoint at a configurable interval.
FR-02	The system shall store all incoming sensor readings in a PostgreSQL relational database with timestamp, field ID, and device ID.
FR-03	The system shall analyze soil parameter values using the trained fertilization decision model to determine optimal fertilizer timing and quantity.
FR-04	The system shall generate alerts when soil conditions fall outside the predefined fertilization threshold ranges for the Yala season.
FR-05	The system shall display real-time soil data, alerts, and historical trends on the Progressive Web App dashboard.
FR-06	The system shall allow administrators to configure threshold values for soil parameters based on crop stage and season.
FR-07	The system shall provide alternative fertilizer recommendations when standard fertilizer availability is marked as limited in the system settings.

5.3 Non- Functional requirements

Performance (FR-01, FR-03, FR-04, FR-05)

The system shall ensure high performance by processing incoming IoT sensor data and generating fertilization alerts within a maximum of 5 seconds after receiving complete data inputs. Real-time data transmission from field devices (FR-01) must support frequent updates without delay, even under multiple field deployments. The fertilization decision engine (FR-03) and alert generation (FR-04) must handle concurrent data streams efficiently. Additionally, the dashboard (FR-05) should load and refresh data smoothly with minimal latency, ensuring farmers receive timely recommendations. The system must maintain consistent response times even when monitoring multiple paddy fields simultaneously.

Accuracy

The system shall maintain high accuracy in both data collection and fertilization recommendations. Sensor data received from IoT devices (FR-01) must be validated and preprocessed to minimize noise and inconsistencies. The fertilization decision model (FR-03) should achieve a minimum of 85% accuracy in predicting optimal fertilizer timing and quantity based on soil parameters and Yala season guidelines. Alerts generated (FR-04) must reflect true soil conditions and avoid false positives or negatives. Continuous calibration of sensors and periodic model retraining using updated datasets will ensure that recommendations remain reliable and scientifically valid for Sri Lankan paddy cultivation conditions.

Reliability

The system shall operate reliably with minimal downtime, ensuring continuous monitoring of soil conditions throughout the cultivation period. IoT devices must consistently transmit data (FR-01), and the backend system should store all readings securely without data loss (FR-02). The dashboard (FR-05) must remain accessible and functional at all times, especially during critical fertilization periods. The system should achieve at least 95% uptime during the Yala season. Fault tolerance mechanisms such as retry logic for data transmission and backup storage must be implemented to handle connectivity failures or sensor interruptions, ensuring uninterrupted service for farmers.

Usability

The system shall be designed with high usability to accommodate farmers with varying levels of digital literacy. The dashboard (FR-05) must present soil data and alerts in a clear, intuitive format using simple visualizations and minimal technical jargon. Administrative controls (FR-06) should allow easy configuration of thresholds without requiring advanced technical knowledge. The system should also clearly present alternative fertilizer recommendations (FR-07) in understandable terms. Support for local languages such as Sinhala and Tamil, along with mobile-friendly design, ensures accessibility. The interface must enable users to interpret alerts and take action quickly without requiring extensive training.

Security

The system shall ensure strong data security and privacy protection for all users and system components. All data transmission from IoT devices (FR-01) must use secure communication protocols such as HTTPS or encrypted channels to prevent unauthorized access. Data stored in the database (FR-02) must be protected using encryption and access control mechanisms. User authentication for the dashboard (FR-05) must be enforced through secure login systems to prevent unauthorized usage. Additionally, the system should implement role-based access control for administrators and users, ensuring that sensitive system configurations and data are only accessible to authorized personnel.

6 Commercialization

The primary stakeholders for the IoT-based paddy fertilization system include small and medium-scale Sri Lankan paddy farmers, agricultural input suppliers, local agronomists, and government agricultural agencies. Farmers are the end-users benefiting from optimized fertilizer usage, while suppliers and agronomists gain insights into demand and soil health trends. Government agencies can leverage the system to enhance national food security and provide policy support for sustainable farming practices.

The product's value proposition for farmers focuses on reducing costs and increasing yield by providing timely alerts for optimal fertilizer application. By continuously monitoring soil conditions and giving actionable recommendations, the system creates tangible financial and operational benefits. For suppliers, it offers accurate data on fertilizer consumption patterns, supporting better inventory planning. Agronomists and extension services gain a scalable platform for monitoring soil health and advising farmers more effectively.

The system addresses both functional and economic dimensions for farmers. Functionally, it provides real-time monitoring, alerts, and fertilizer optimization. Economically, it reduces wastage and increases profitability. For suppliers and government stakeholders, the product enhances operational efficiency and decision-making capabilities, creating a compelling value proposition that aligns customer needs with the unique features of the IoT component.

The commercialization strategy includes initial pilot testing in selected Yala season fields to demonstrate cost savings and yield improvements. Feedback will be collected to refine alerts, dashboard usability, and sensor placement. The system can then be offered through subscription models, bundled packages with sensors, or through partnerships with local agricultural cooperatives, ensuring accessibility and affordability for small-scale farmers.

Long-term commercialization can involve integration with mobile platforms, providing analytics dashboards, and offering predictive models for crop planning. Partnerships with fertilizer companies and government agencies can expand adoption while enabling local recommendations. By demonstrating measurable benefits and scalability, the solution can transition from pilot deployment to widespread adoption across Sri Lanka's paddy farming regions.

7 BUDGET AND BUDGET JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Table 1 --- Budget Table

Description	Cost
Design	15,000
Development	25,000
Testing	10,000
Deployment	15,000
Maintainance	10,000
Total	**75,000**

8 WORK BREAKDOWN CHART

WP	Work Package	Key Tasks	Deliverable
WP1	Data Collection & IoT Device Design (Week 1–4)	Select soil parameters (moisture, temperature, pH, EC, humidity); design IoT device using ESP32; integrate sensors; plan installation areas in paddy field; begin collecting sample soil data.	Working IoT prototype; sensor integration setup; initial soil dataset.
WP2	Field Deployment & Data Transmission (Week 5–8)	Install IoT devices in multiple crop zones; configure WiFi/LoRa communication; develop REST API; send sensor data to server; verify real-time data collection.	Active field devices; validated data transmission; API data pipeline.
WP3	Backend, Database & Data Processing (Week 9–12)	Develop backend server; create PostgreSQL database; store sensor readings; implement preprocessing scripts; prepare dataset for fertilization decision model.	Functional backend; database schema; processed soil dataset.
WP4	Fertilization Decision Model & Alert System (Week 13–18)	Define fertilization thresholds for Yala season; train rule-based / ML model; generate alerts based on soil values; implement alternative fertilizer logic; test prediction accuracy.	Fertilization model; alert generation module; validation report.
WP5	Dashboard, Deployment & Testing (Week 19–24)	Develop Next.js dashboard; integrate API; deploy on Hostinger VPS; field testing with farmers; evaluate performance; fix errors; finalize system.	Production-ready dashboard; deployed system; final tested component.

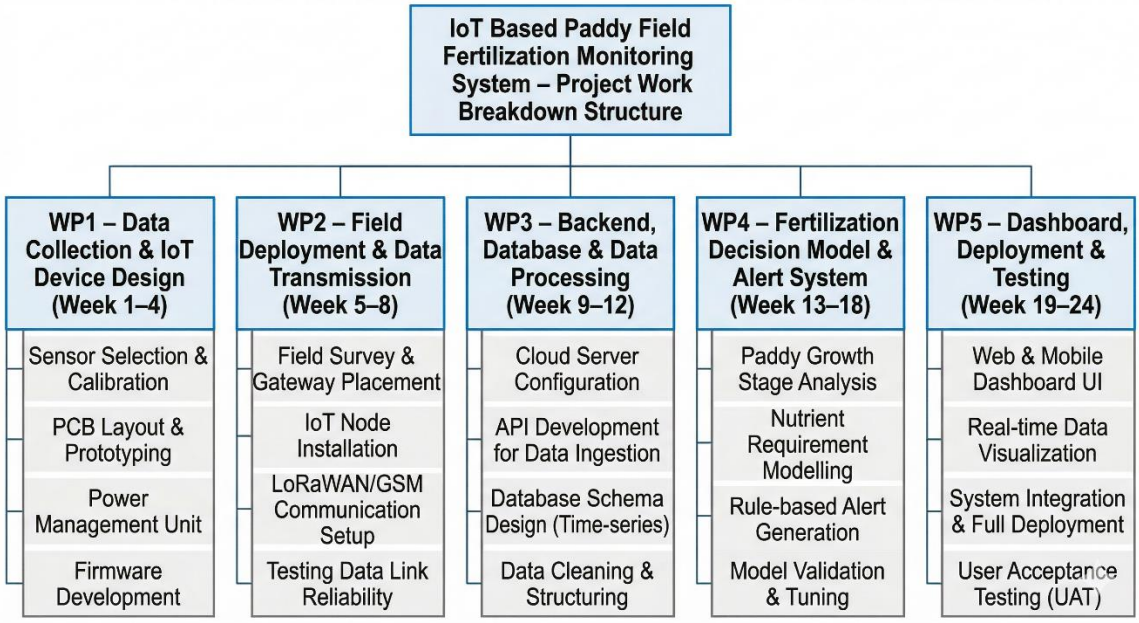


Figure 3 -- Work breakdown Chart

9 GANTT CHART

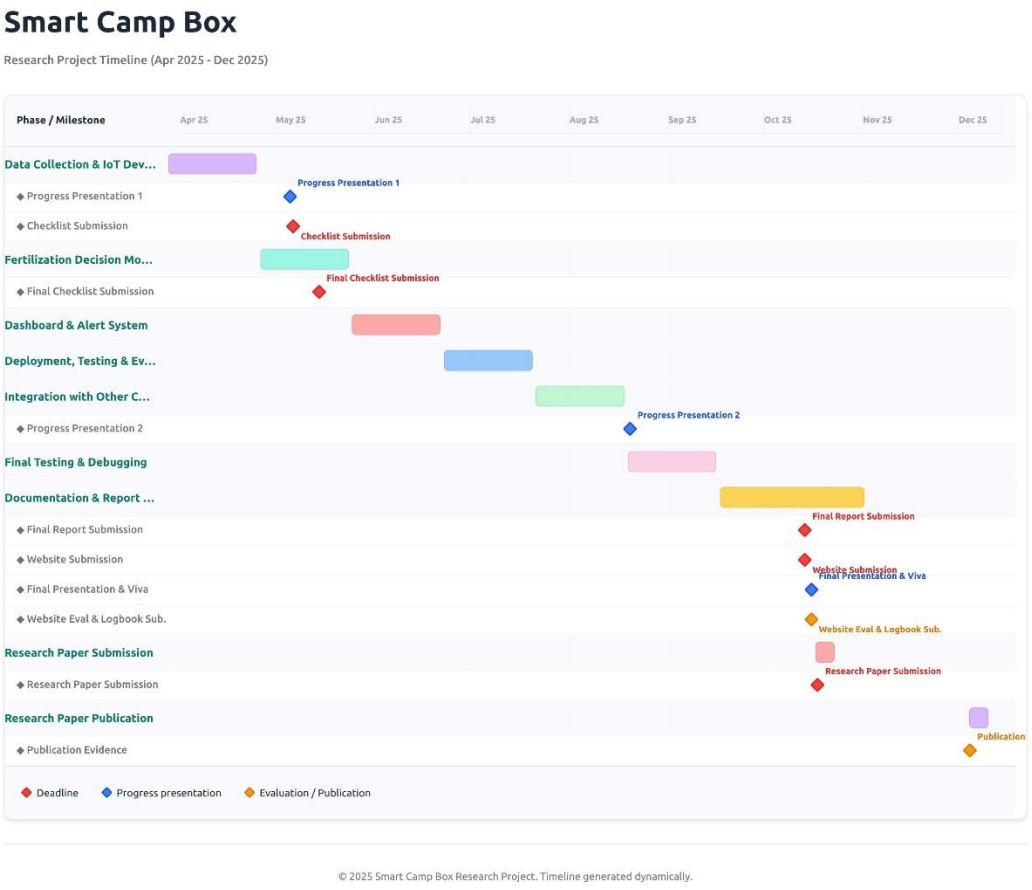


Figure 2 -- Gannt Chart

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11 APPENDICES

Appendix A - Soil Sensor Reference Values

Source: Department of Agriculture Sri Lanka (DOA), 2022. Used as baseline thresholds for SO2 fertilization decision engine.

Parameter	Optimal Range (Yala Season)	Critical Thresholds	Notes
Soil Moisture %	60–80%	<40% / >90%	Low moisture triggers irrigation alert
Soil pH	5.5–6.5	<5 / >7	Outside range reduces nutrient uptake
N (mg/kg)	25–35	<15 / >50	Used for nitrogen fertilization guidance

P (mg/kg)	15–25	<10 / >40	Phosphorus application recommended
K (mg/kg)	120–200	<80 / >300	Potassium dose adjustments calculated
Temperature °C	28–32	<20 / >38	Extreme temps affect growth
Humidity %	70–85	<50 / >95	Used for evapotranspiration adjustment

Appendix B - Fertilization Dosage Reference Table

Source: DOA Sri Lanka, Yala Season Guidelines, 2022. Used as training and validation input for SO2 and SO4.

Growth Stage	N (kg/ha)	P (kg/ha)	K (kg/ha)	Remarks
Germination	20	10	15	Initial basal application
Tillering	25	12	20	Split application recommended
Panicle Initiation	20	10	15	Adjust based on soil reading
Heading	15	5	10	Optional top-dressing
Grain Filling	10	0	5	Only if nutrient deficiency

Appendix C - Dataset Sources and Access Details

Dataset	Source	Access Method	Used For
Real-time IoT soil data	IT22178718 (group module)	REST API (internal)	SO1 live model input
Current fertilizer/market cost	DOA Sri Lanka / Hector Kobbekaduwa Inst.	Institutional request	SO3 revenue and cost projections
Historical district yield data	DCS Sri Lanka	Free download: statistics.gov.lk	SO1 base yield model training
Fertilization guidelines	DOA Yala season, 2022	PDF download	SO2 training & decision rules

Appendix D — Risk Analysis

Risk	Level	Impact	Mitigation
IoT sensor malfunction or misreading	Medium	Incorrect fertilization recommendation	Implement redundancy, periodic calibration
API unavailability	Medium	SO1/SO2 blocked	Develop mock API, pre-fill historical data

		without live inputs	
Fertilizer supply shortage due to war	High	Recommendations cannot be applied in-field	Include alternative organic or local fertilizers
Extreme Yala season heat (>38°C)	Low	Reduced crop response to fertilization	Adjust dose thresholds, include irrigation alerts
Farmer low digital literacy	Low	Usability metrics below target	Sinhala guided walkthroughs, video tutorials

Appendix E — Ethical Considerations

- All external datasets (DCS, IRRI SES) are publicly available through official channels. No personal data is involved in obtaining these datasets.
- Farmer participation in usability testing is entirely voluntary. Written informed consent will be obtained before any testing session.
- No personally identifiable information (name, NIC, location) will be stored in the production system database. Farmer profiles use anonymized IDs.
- SLIIT Research Ethics Committee approval will be obtained prior to conducting any primary data collection or farmer usability testing involving human participants.
- AI writing assistance tools were used to support drafting and structuring of this report. All research direction, component design decisions, novelty justification, and analytical interpretation are the original work of the student.